

Logistic Growth

Where a limited population grows fastest, why that place is not the top of the curve, and what it takes to push a population past K .

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/logistic-growth/

1 Before you open anything

Answer from what you already know. Being right now is not the point — having something on paper to argue with later is.

PREDICTION

A population grows logistically toward a carrying capacity of $K = 1,000$ individuals. At which of these sizes does it add the *most* new individuals per day: $N = 100$, $N = 500$, $N = 900$, or $N = 1,000$? Circle one and write the number.

Write one sentence saying what reasoning led you there.

2 Reading one population two ways

Open the demonstration. The upper plot is N against time; the lower plot is dN/dt against N . The panels on the right report both in numbers.

1. Press **Reset all**. The sliders should now read $r = 0.50 \text{ d}^{-1}$, $K = 1,000$, $N_0 = 10$, **Time window** = 60 d and **Lag** $\tau = 0.0 \text{ d}$.
2. Drag the **Day shown** slider to each day in the table. Clicking the slider once and then using the arrow keys gets you to an exact value.
3. For each day, copy the four values out of the panel headed **At day ...**

Day shown	Logistic N	Its growth dN/dt	Exponential N	Exponential $\div$ logistic
0.0				
5.0				
10.0				
15.0				
20.0				
30.0				

a. Between which two rows is *Its growth* dN/dt at its largest? Give the logistic N at that row, and compare it with the value in the readout labelled **Population there, $K/2$** .

b. From day 20 to day 30 the logistic population barely changes, but *Exponential $\div$ logistic* changes enormously. Say what each of those two facts tells you, in one sentence each.

c. Record the four values in the panel **Where growth is fastest**:

Notice the units. r is a rate *per individual per day* (d^{-1}), while dN/dt is individuals per day for the whole population. They are not the same quantity and are never interchangeable in a calculation.

3 Where the peak sits, and how high it is

Change one thing at a time. Leave $r = 0.50 d^{-1}$ fixed for the whole table and move only **Carrying capacity K**. Both readouts you need are in the **Where growth is fastest** panel and neither depends on the day shown.

K	Population there, $K/2$	Fastest growth $rK/4$	Growth at $N = 0.9K$
200			
600			
1,000			
1,400			
2,000			

a. K doubles from 1,000 to 2,000. By what factor does the fastest growth rate change? By what factor does the population size at which it happens change?

b. Now set **K = 1,000** and move **r** to **1.00 d⁻¹**. Record both readouts again. Which one moved, and which one did not?

c. State the rule in your own words: for any r and any K, the growth rate of a logistic population is greatest when N is _____ and the value there is _____.

d. Go back to your prediction in part 1. Was it right? If it was not, name the specific assumption behind it that turned out not to hold.

4 Working backwards, and one rule that breaks

Exam questions rarely hand you r and K and ask for the curve. More often they hand you something measured and ask for r or K .

1. A population is observed to grow fastest at **90 individuals per day**, and that fastest growth happens when the population is **400**. Work out K and r on paper first, using $N_{\text{peak}} = K/2$ and $(dN/dt)_{\text{max}} = rK/4$.
 2. Now set the two sliders to your answers and check the **Where growth is fastest** panel.
- a. Your working, then the two readouts as the demonstration shows them:

Everything above rests on a rule that sounds safe: *a population cannot exceed its carrying capacity*. Test it.

1. Press the **No lag** button, then **Damped**, then **Sustained cycles**. After each, record the two readouts in the lag panel.
2. With **Damped** selected, set **Day shown** to **12.0 d** and read the tile **With the lag: N and dN/dt**.

Button	Lag τ	$r \times \tau$	Highest population reached, as a share of K
No lag			
Damped			
Sustained cycles			

- b. At day 12.0 with the damped lag, is the population above K ? Is it still growing? Copy the sentence the tile gives you.

- c. In $dN/dt = rN(t)[1 - N(t-\tau)/K]$, say which N is being counted in each of the two factors, and use that to explain how the population can sit above K and still be adding individuals.

d. Look at the lower plot with the lag on. The coloured path no longer lies on the parabola. What does that tell you about whether dN/dt can be worked out from today's N alone?

Overshoot in this model comes from the delay and from nothing else. Read the **Honest limits** panel: a real crash can also come from K itself dropping — a drought or a disease can halve K in a season while the population is still sitting at the old number.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, check yourself, and where you were wrong write down which step went off.

1. A population has $r = 0.60 \text{ d}^{-1}$ and $K = 1,500$ individuals. Find the population size at which it grows fastest, and the growth rate there. Give units.

2. A population has $r = 0.25 \text{ d}^{-1}$ and $K = 2,000$. Find dN/dt when $N = 600$. Then, without doing the arithmetic again, name a *different* population size with exactly the same dN/dt , and justify it in one line.

3. Fisheries are often managed for maximum sustainable yield at roughly half the unfished stock. Using the two competing factors in $rN(1 - N/K)$, explain why a stock held near $K/2$ can replace itself faster than one left near K . No calculation. Then name one thing this model leaves out that a real fishery would have to account for.

4. A population has $r = 0.50 \text{ d}^{-1}$ and a reproductive lag of $\tau = 0.6$ days. Compute $r\tau$ and predict, using the thresholds $1/e = 0.368$ and $\pi/2 = 1.571$, whether the population overshoots K . Repeat for $\tau = 3.5$ days.

6 On your own

Nothing here is graded, and the demonstration does more than this worksheet uses.

- Set $K = 1,000$ with the lag at 0 and push **Starting population N_0** up to 1,400. Which direction does the curve run now, and what does that say about calling K a ceiling?
 - Widen the **Time window** to 200 days and watch *Exponential ÷ logistic*, plus the label saying which day the exponential leaves the chart. How large does that ratio get, and what does it say about how long an exponential model stays usable?
 - Turn a lag on, press **Play**, and follow the moving dot on the lower plot rather than the upper one. Where in the loop is the population above K and still climbing, and where is it below K and still falling?
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